👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 5 Carburetion & Fuel (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Venturi Lab**: Title only in Canvas; our lab is team-written.
- **Carburetor Article**: Scanned image, source unknown; our reading replaces it.
- **Carburetor Quiz key**: The paper quiz has no key in the export; our key is team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⛽ Carburetion & Fuel: Guided Notes

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

Use the class slides and your reading. Key words:

carburetor *venturi* *air-fuel ratio* *jet* *float* *choke* *throttle* *diaphragm* *ethanol* *stabilizer* *rich* *lean*

A. Venturi

1. Explain how the venturi pulls fuel into the air.

B. Parts

C. Mixture

3. Too much fuel is a _____ mixture.

4. Too much air is a _____ mixture.

5. 2-stroke fuel at 50:1 means…

A. 50 oil to 1 gas

B. 50 parts gas to 1 part oil

C. Equal parts

⛽ Carburetion & Fuel: Reading

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

How a Carburetor Works

An engine needs the right mix of air and fuel to run, about 14.7 parts air to 1 part fuel by weight. The carburetor makes this mix, turns liquid fuel into a fine mist, and controls engine speed.

The key idea is the venturi, a narrow section in the carburetor's throat. As air rushes through the narrow spot, it speeds up and its pressure drops. Normal air pressure on the fuel in the bowl then pushes fuel up through a small jet and into the fast airstream, where it mixes with the air.

Other parts control the mix. The float and needle keep the fuel in the bowl at the correct level. The throttle plate controls how much mixture enters the engine, changing speed. The choke limits air when the engine is cold so the mixture is richer. Float carburetors are used on mowers; diaphragm carburetors, used on chainsaws and trimmers, pump fuel with a flexible diaphragm so they work at any angle.

Fuel quality matters. Most gas contains ethanol, which absorbs water and can leave gum that clogs tiny jets. Use fresh fuel, add stabilizer for storage, and mix two-stroke oil at the correct ratio.

Check your reading

1. Why is the venturi narrow?

2. What does the choke do?

3. Why can ethanol cause problems?

⏳ Venturi & Carburetor Lab: Lab Sheet

Small Engines Overhaul · Unit 5 · Level 3 · goes with Venturi Lab + Carburetor Handout

Name: _____

Question: How does fast-moving air pull fuel into a carburetor?

Materials

2 straws, cup of water, paper strip, practice carburetor

⚠️ Safety

- Safety glasses
- Staff handle carb cleaner in ventilation
- Tools with adult supervision

Prediction

What will happen to the water when you blow across the straw?

_____

_____

Procedure

1. Blow across paper; observe.
2. Blow across straw in water; observe.
3. Identify venturi, jet, choke, throttle on a carburetor.
4. Remove bowl; identify float and needle.
5. Reassemble; test linkage.

Data

Test	What happened	Why
Paper strip		
Straw in water		

Analysis (Claim, Evidence, Reasoning)

1. Claim: how does a carburetor draw fuel? Evidence: your two tests. Reasoning: connect to the venturi.

⛽ Carburetion & Fuel: Quick Check

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The ideal air-fuel ratio is about…

A. 1:14.7 | B. 50:1 | C. 14.7:1

2. Which carburetor works at any angle?

A. Diaphragm | B. Float | C. Gravity

3. Closing the choke makes the mixture…

A. Leaner | B. Unchanged | C. Richer

4. A stuck float can cause…

A. Flooding | B. Low compression | C. No spark

5. Explain the venturi effect.

6. Give two ways to keep fuel from causing carburetor problems.

⛽ Carburetion & Fuel: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

Use the class slides and your reading. Key words:

carburetor · *venturi* · *air-fuel ratio* · *jet* · *float* · *choke* · *throttle* · *diaphragm* · *ethanol* · *stabilizer* · *rich* · *lean*

A. Venturi

1. Explain how the venturi pulls fuel into the air.
Air speeds up in the narrow throat, pressure drops, and higher pressure on the bowl pushes fuel up the jet.

B. Parts

C. Mixture

3. Too much fuel is a **rich** mixture.

4. Too much air is a **lean** mixture.

5. 2-stroke fuel at 50:1 means…

A. 50 oil to 1 gas | **B. 50 parts gas to 1 part oil** | C. Equal parts

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

⏳ Venturi & Carburetor Lab: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 5 · Level 3 · goes with Venturi Lab + Carburetor Handout

Name: _____

Question: How does fast-moving air pull fuel into a carburetor?

Materials

2 straws, cup of water, paper strip, practice carburetor

⚠️ Safety

- Safety glasses
- Staff handle carb cleaner in ventilation
- Tools with adult supervision

Prediction

What will happen to the water when you blow across the straw?

It will rise in the straw.

Procedure

1. Blow across paper; observe.
2. Blow across straw in water; observe.
3. Identify venturi, jet, choke, throttle on a carburetor.
4. Remove bowl; identify float and needle.
5. Reassemble; test linkage.

Data

Test	What happened	Why
Paper strip	**lifted up**	**fast air above, lower pressure**
Straw in water	**water rose**	**low pressure at straw top**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: how does a carburetor draw fuel? Evidence: your two tests. Reasoning: connect to the venturi.
Fast air in the venturi lowers pressure; bowl pressure pushes fuel up the jet like water up the straw.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

⛽ Carburetion & Fuel: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 5 · Level 3 (closer to the class work) · goes with the Carburetor Handout, video handout, quiz and Venturi Lab

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The ideal air-fuel ratio is about…

A. 1:14.7 | B. 50:1 | **C. 14.7:1**

2. Which carburetor works at any angle?

A. Diaphragm | B. Float | C. Gravity

3. Closing the choke makes the mixture…

A. Leaner | B. Unchanged | **C. Richer**

4. A stuck float can cause…

A. Flooding | B. Low compression | C. No spark

5. Explain the venturi effect.

Narrow throat speeds air, lowering pressure; fuel is pushed into the airstream.

6. Give two ways to keep fuel from causing carburetor problems.

Use fresh fuel, add stabilizer, run carb dry for storage, correct 2-stroke mix.

🔑 *Written by our team from his unit calendar, worksheets and review games, please check.*